

Module 1: Basic programming in R

Benjamin Smith

07/06/2026

Methods and computing camp

This summer we will together learn and review materials of statistical computing and methods.

What do we do during lectures?

Materials will be available at course website. Lecture notes are created by Rmarkdown.

- **If you have questions, feel free to interrupt or send a message in the chat.**

We will cover 5 modules of statistical methods and computing.

- Each module takes ~2.5 hours.

What contents we will cover?

Module	Topics	References
1	R, Rstudio, and Rmarkdown; data wrangling and graphing; LaTeX; Elementary data analysis	-
2	Statistical inference (I)	AoS Chp 1-5; AoS Chp 6; C&B Chp 6.3, 7
3	Statistical inference (II)	AoS Chp 8; C&B Chp 8
4	Linear Regression & Generalized Linear Models	AoS Chp 13; C&B Chp 11-12
5	Simulation and parallel computing; Bootstrap	C&B Chap 10; AoS Chp 5, 24;

What contents we will not cover?

- Nonparametric inference (STA3000)
- Bayesian inference (STA3000, STA2201)
- Computing of Bayesian inference (STA2201)
- Shiny and blogdown in R

Exercises!

- Available for each module.
- Exercises can be difficult.
- Group discussion is recommended.

Module 1: Basic programming in R

We will review R, Rstudio, and Syntax of R together.

- Rstudio (Knitr)
- Basic data types
- Basic data structures
- Functions
- For loops

Useful resources:

- Tidyverse style guide
- The R Inferno

Introduction to R and Rstudio

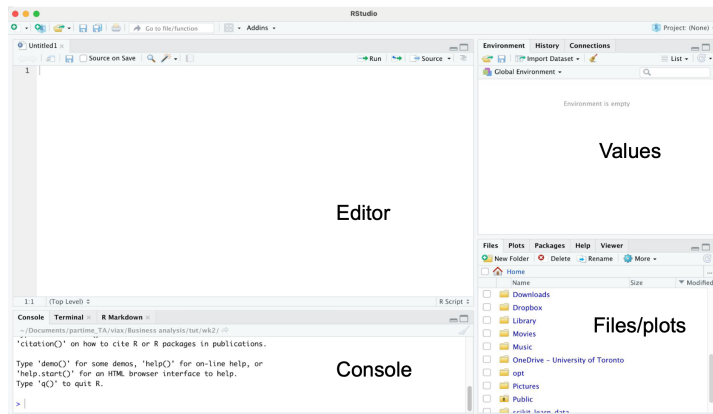
R is a free statistical software. We use R frequently/intensively during our study.

First please download R and its IDE Rstudio (if you haven't).

- <https://www.r-project.org/>
- <https://www.rstudio.com/>

Studio

- Editor: edit or save the file.
- Console: outputs.
- Values: store values of assigned.
- Files/plots/packages/help.



How to set working directory?

Working directory is important since you might want to read and import data from other files. It is recommended to put these files under the same directory with your scripts.

- Method 1: Session -> set working directory.
- Method 2: Files -> Navigate to your directory.
- Method 3: `setwd()`.

Alternative: - Make an Project - Files -> New Project; - This will create an .Rproj file

How to install packages?

Common packages in statistical analysis with R:

- tidyverse/dplyr
- ggplot
- kableExtra or gridExtra

Several options to install packages:

- Method 1: Tools -> install packages
- Method 2: Packages window.
- **Method 3:** `install.packages()`
- Method 4: `devtools::install_github()`
- Method 5: `pak::pkg_install()`

Use Method 4 if you want to use a package on github. Use Method 5 if you are encountering dependency issues.

Ready with your Rstudio?

Let's code!

Vector

Can contain numerical, string, or Boolean value.

Store data = make an assignment. `c()` is for component.

```
v <- c(TRUE, FALSE, TRUE, TRUE, FALSE)
v <- c("python", "mathlab", "R")
v <- c(6, 5, 4, 3, 2, 1)
```

Are you familiar with the output of these?

```
which(v == 3)
```

```
## [1] 4
```

```
v[2]
v[-2]
v[2:3]
v[v < 4]
which(v == 3)
```

Matrix

```
mymat <- matrix(c(1:10), nrow = 2, ncol = 5,  
                byrow = TRUE)  
  
mymat  
mymat[1, 5]  
mymat[2, ]  
mymat[c(1:2), c(1:2)]
```

Data frame

```
studentID <- c(1, 2, 3, 4)
age <- c(17, 18, 16, 19)
gender <- c("M", "F", "M", "M")
studentData <- data.frame(studentID, age, gender)
```

```
rownames(studentData) <- c("A", "B", "C", "D")
```

```
colnames(studentData) <- c("ID", "age", "gender")
studentData
```

```
##   ID age gender
## A  1  17      M
## B  2  18      F
## C  3  16      M
## D  4  19      M
```

Linear Regression

$$y_i = \beta^T x_i + \epsilon_i$$

```
x <- c(1:5)
eps <- rnorm(5)
y <- 2*x + eps
mod <- lm(y ~ x)
```


Linear Regression

```
summary(mod)
```

```
##
## Call:
## lm(formula = y ~ x)
##
## Residuals:
##      1      2      3      4      5
## -0.0519 -0.2327  0.1781  0.5494 -0.4429
##
## Coefficients:
##              Estimate Std. Error t value Pr(>|t|)
## (Intercept)   1.0238     0.4638   2.208  0.11436
## x             1.6181     0.1398  11.572  0.00139 **
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 0.4422 on 3 degrees of freedom
## Multiple R-squared:  0.9781, Adjusted R-squared:  0.9708
## F-statistic: 133.9 on 1 and 3 DF, p-value: 0.001386
```

List

```
g <- "My List"
h <- c(2, 3, 5, 7)
j <- matrix(1:10, nrow = 5, byrow = FALSE)
k <- c("one", "two", "three")
mylist <- list(title = g, ages = h, j, k)
```

```
mylist[[2]]
```

```
## [1] 2 3 5 7
```

```
mylist$ages
```

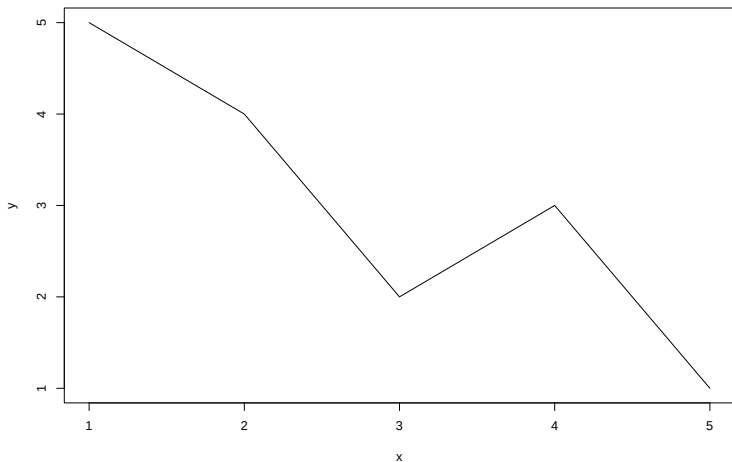
```
## [1] 2 3 5 7
```

```
mylist[[1]]
```

```
## [1] "My List"
```

Plotting

```
y <- c(5, 4, 2, 3, 1)
x <- c(1, 2, 3, 4, 5)
plot(x, y, type = "l")
```



Other

Important for stats research.

- `floor(v)`, `ceiling(v)`
- `round(v, 2)`
- `rnorm()`, `rexp()`, `rbinom()`, etc generate random variables
- Functions

```
rnorm(5, mean = 0, sd = 1) # 5 random numbers from  $N(0, 1)$ 
```

```
## [1] -2.028852499 0.550717171 -0.095986230 -0.266392983 -0.005074615
```

```
round(2.333333, 2)
```

```
## [1] 2.33
```

```
func <- function(x1, x2) {  
  y <- 3 * x1^2 + 3 * x2^2 - 2 * x1^2 * x2  
  return(y)  
}  
func(1, 2)
```

```
## [1] 11
```

If else

What is the output?

```
p <- 3
if (p <= 2) {
  print("p <= 2!")
} else {
  print("p > 2!")
}
```

for loop

```
v <- c(1, 2, 4, 3)
w <- c(0, 0, 0, 0)
t <- 0
```

```
for (i in v) {
  t <- t + 1
  w[t] <- w[t] + i
  print(w)
}
```

```
## [1] 1 0 0 0
```

```
## [1] 1 2 0 0
```

```
## [1] 1 2 4 0
```

```
## [1] 1 2 4 3
```

```
w
```

```
## [1] 1 2 4 3
```

while loop

```
i <- 1
while (i <= 10) {
  print(i)
  i <- i + 1
}
```

```
## [1] 1
## [1] 2
## [1] 3
## [1] 4
## [1] 5
## [1] 6
## [1] 7
## [1] 8
## [1] 9
## [1] 10
```

next

```
alphabet <- LETTERS[1:6]
for (i in alphabet) {
  if(i == 'D') {
    next
  }
  print(i)
}
```

```
## [1] "A"
## [1] "B"
## [1] "C"
## [1] "E"
## [1] "F"
```


break

```
alphabet <- LETTERS[7:12]
for (i in alphabet) {
  if (i == 'K') {
    break
  }
  print(i)
}
```

```
## [1] "G"
## [1] "H"
## [1] "I"
## [1] "J"
```

apply

It can be applied on matrix, vector, data frame, and loop through row or column (defined by the second input - MARGIN).

Syntax: `apply(X, MARGIN, FUN, ...)`

```
f <- function(x) {  
  ts <- 2*x^2  
  return(ts)  
}  
  
ii <- matrix(1:4, nrow = 1)  
apply(ii, 1, f)
```

```
##      [,1]  
## [1,]    2  
## [2,]    8  
## [3,]   18  
## [4,]   32
```

Knit in Rstudio

Common types of R files: .R, .Rmd, .qmd

- Simulations, e.g. for loops, functions, I use .R
- Reporting, analysis, plotting, I use .Rmd or .qmd

.Rmd/.qmd files can be converted to pdf, html through knitr.

- yaml style.

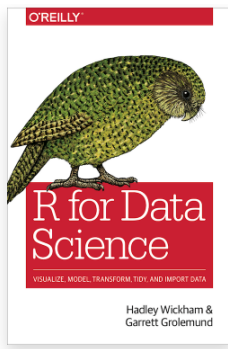
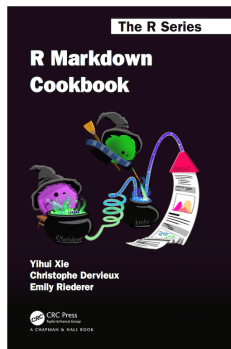
```
---  
title: 'Module 1: Basic programming in R'  
date: "04/15/2022"  
output:  
  beamer_presentation  
---
```

More yaml style

```
---  
title: "A summary of xx"  
date: "02/14/2022"  
output:  
  pdf_document:  
    toc: true  
    number_sections: true  
---
```

Resources

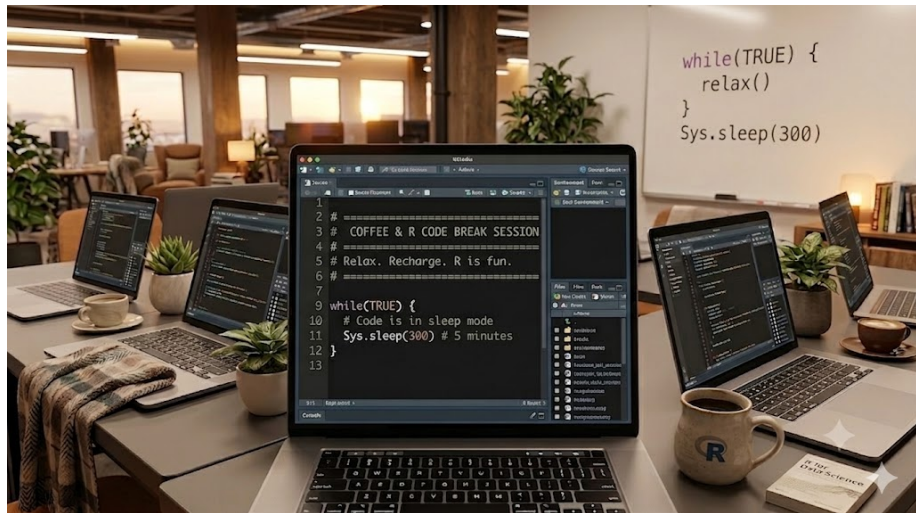
- “R Markdown Cookbook” by Yihui Xie.
 - <https://bookdown.org/yihui/rmarkdown-cookbook/>
- “R for Data Science (2e)” by Hadley Wickham
 - <https://r4ds.hadley.nz/quarto.html>



Code style

- Tidyverse style guide
- `styler` software embedded in Rstudio - allows you to interactively restyle selected text, files, or entire projects.
- `lintr` software embedded in Rstudio - performs automated checks to confirm that you conform to the style guide.

Break



Outline

We will review R, Rstudio, and Syntax of R together.

- LaTeX/Markdown
- Tidy data, processing (tidyverse)
- Graphing (ggplot2)

LaTeX and Markdown

LaTeX is useful for documents with mathematical formulas.

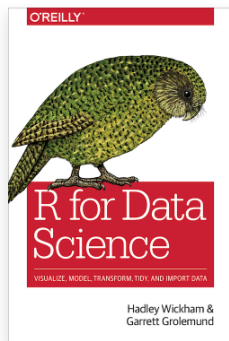
- Overleaf - an online, collaborative LaTeX editor
- LaTeX mathematical symbols
- Inline equation e.g. (`\alpha`) returns α
- Equation e.g. (`e = mc^2`) returns

$$e = mc^2$$

Markdown is appealing for formatting, e.g. headings, bold text, text with codes, ...

Resources

- “R for Data Science: Import, Tidy, Transform, Visualize, and Model Data” by Hadley Wickham.
- Brendan R. E. Ansell’s “Introduction to R - tidyverse” [[link](#)]



Data import

```
df <- read.table("mtcars.txt", header = TRUE)
head(df) # Show the first 6 rows.
```

```
##   Cntry lper100k weight length
## 1    US      19.8   2178   5.92
## 2  Japan      9.9   1026   4.32
## 3    US     10.8   1188   4.27
## 4    US     12.5   1444   5.11
## 5    US     12.5   1485   5.03
## 6    US     12.5   1485   5.03
```

Other options

CSV files.

- `base::read.csv()` in the base `r`.
- `readr::read.csv()` in “`readr`” package (much faster).
- `data.table::fread()` in “`data.table`” package (much more faster).

Rdata.

- `load()` in the base `r`.

Tidy data

The goal is to clean the dataset so it is much easier to use.

Specifically,

- Each variable must have its own column.
- Each observation must have its own row.
- Each value must have its own cell.

We will focus on the functions from “tidyverse” package.

```
library(tidyverse)
```

```
## Warning: package 'ggplot2' was built under R version 4.4.3
```

Tidy data 1: pivoting

For a dataset having column names are not names of variables, but values of a variable, e.g.

```
# table4a documents the number  
# of documented TB cases in  
# Afghanistan, Brazil and China  
table4a
```

```
## # A tibble: 3 x 3  
##   country    `1999` `2000`  
##   <chr>      <dbl> <dbl>  
## 1 Afghanistan    745   2666  
## 2 Brazil        37737  80488  
## 3 China         212258 213766
```

- Need to change 1999, 2000 to a column? named as "year".
- Need to change the values of 1999, 2000 as "cases".

We can use `pivot_longer()` from the "tidyverse" package.

Pivot longer

```
table4a %>%  
  pivot_longer(c(`1999`, `2000`),  
    names_to = "year", values_to = "cases")
```

```
## # A tibble: 6 x 3  
##   country    year cases  
##   <chr>      <chr> <dbl>  
## 1 Afghanistan 1999    745  
## 2 Afghanistan 2000   2666  
## 3 Brazil      1999  37737  
## 4 Brazil      2000  80488  
## 5 China       1999 212258  
## 6 China       2000 213766
```

Another example

```
table2 %>% head(5)
```

```
## # A tibble: 5 x 4
##   country      year type      count
##   <chr>      <dbl> <chr>    <dbl>
## 1 Afghanistan 1999 cases      745
## 2 Afghanistan 1999 population 19987071
## 3 Afghanistan 2000 cases      2666
## 4 Afghanistan 2000 population 20595360
## 5 Brazil      1999 cases      37737
```

- case and population are two variables and should be converted into columns.

We can use `pivot_wider()`.

Pivot wider

```
table2 %>%  
  pivot_wider(names_from = type, values_from = count)
```

```
## # A tibble: 6 x 4  
##   country      year  cases population  
##   <chr>      <dbl> <dbl>      <dbl>  
## 1 Afghanistan 1999     745   19987071  
## 2 Afghanistan 2000    2666  20595360  
## 3 Brazil      1999   37737  172006362  
## 4 Brazil      2000   80488  174504898  
## 5 China       1999 212258 1272915272  
## 6 China       2000 213766 1280428583
```

Transform data

Use the “pipes” from the “tidyverse” package, a powerful tool for clearly expressing a sequence of multiple operations, with the combination of the following functions:

- `select()`
- `filter()`
- `arrange()`
- `mutate()`
- `summarise()`
- `group_by()`

Dataset - Diamonds

A dataset containing the prices and other attributes of almost 54,000 diamonds.

```
head(diamonds)
```

```
## # A tibble: 6 x 10
##   carat cut      color clarity depth table price      x      y      z
##   <dbl> <ord>    <ord> <ord>    <dbl> <dbl> <int> <dbl> <dbl> <dbl>
## 1  0.23 Ideal      E     SI2     61.5   55   326   3.95   3.98   2.43
## 2  0.21 Premium    E     SI1     59.8   61   326   3.89   3.84   2.31
## 3  0.23 Good       E     VS1     56.9   65   327   4.05   4.07   2.31
## 4  0.29 Premium    I     VS2     62.4   58   334   4.2    4.23   2.63
## 5  0.31 Good       J     SI2     63.3   58   335   4.34   4.35   2.75
## 6  0.24 Very Good J     VVS2     62.8   57   336   3.94   3.96   2.48
```

Select

Use `select()` to get a column, e.g. “color”

```
diamonds %>%  
  select(color) %>%  
  head()
```

```
## # A tibble: 6 x 1  
##   color  
##   <ord>  
## 1 E  
## 2 E  
## 3 E  
## 4 I  
## 5 J  
## 6 J
```

```
# Equivalent to...  
head(diamonds$color)
```

```
## [1] E E E I J J  
## Levels: D < E < F < G < H < I < J
```

Select

Use `select()` to remove a column, e.g. “color”

```
diamonds %>%  
  select(-color)
```

```
## # A tibble: 53,940 x 9  
##   carat cut      clarity depth table price     x     y     z  
##   <dbl> <ord>    <ord>    <dbl> <dbl> <int> <dbl> <dbl> <dbl>  
## 1  0.23 Ideal    SI2      61.5   55   326   3.95   3.98   2.43  
## 2  0.21 Premium SI1      59.8   61   326   3.89   3.84   2.31  
## 3  0.23 Good    VS1      56.9   65   327   4.05   4.07   2.31  
## 4  0.29 Premium VS2      62.4   58   334   4.2    4.23   2.63  
## 5  0.31 Good    SI2      63.3   58   335   4.34   4.35   2.75  
## 6  0.24 Very Good VVS2    62.8   57   336   3.94   3.96   2.48  
## 7  0.24 Very Good VVS1    62.3   57   336   3.95   3.98   2.47  
## 8  0.26 Very Good SI1      61.9   55   337   4.07   4.11   2.53  
## 9  0.22 Fair    VS2      65.1   61   337   3.87   3.78   2.49  
## 10 0.23 Very Good VS1      59.4   61   338   4     4.05   2.39  
## # i 53,930 more rows
```

```
# Need to assign the change to the original dataset, otherwise, the deletion won't affect the dataset.  
diamonds <- diamonds %>%  
  select(-color)
```

Filter

Use `filter()` to filter by some condition, e.g. filter all price > 335

```
diamonds %>%  
  filter(price > 335)
```

```
## # A tibble: 53,935 x 10  
##   carat cut      color clarity depth table price      x      y      z  
##   <dbl> <ord>    <ord> <ord>    <dbl> <dbl> <int> <dbl> <dbl> <dbl>  
## 1  0.24 Very Good J      VVS2    62.8    57    336    3.94    3.96    2.48  
## 2  0.24 Very Good I      VVS1    62.3    57    336    3.95    3.98    2.47  
## 3  0.26 Very Good H      SI1     61.9    55    337    4.07    4.11    2.53  
## 4  0.22 Fair      E      VS2     65.1    61    337    3.87    3.78    2.49  
## 5  0.23 Very Good H      VS1     59.4    61    338    4      4.05    2.39  
## 6  0.3   Good      J      SI1     64     55    339    4.25    4.28    2.73  
## 7  0.23 Ideal      J      VS1     62.8    56    340    3.93    3.9    2.46  
## 8  0.22 Premium    F      SI1     60.4    61    342    3.88    3.84    2.33  
## 9  0.31 Ideal      J      SI2     62.2    54    344    4.35    4.37    2.71  
## 10 0.2   Premium    E      SI2     60.2    62    345    3.79    3.75    2.27  
## # i 53,925 more rows
```

Filters with multiple conditions

```
diamonds %>%  
  filter(price > 335 & depth < 64)
```

```
## # A tibble: 51,849 x 10
```

```
##   carat cut      color clarity depth table price      x      y      z  
##   <dbl> <ord>    <ord> <ord>    <dbl> <dbl> <int> <dbl> <dbl> <dbl>  
## 1  0.24 Very Good J      VVS2    62.8    57    336  3.94  3.96  2.48  
## 2  0.24 Very Good I      VVS1    62.3    57    336  3.95  3.98  2.47  
## 3  0.26 Very Good H      SI1     61.9    55    337  4.07  4.11  2.53  
## 4  0.23 Very Good H      VS1     59.4    61    338   4     4.05  2.39  
## 5  0.23 Ideal    J      VS1     62.8    56    340  3.93  3.9   2.46  
## 6  0.22 Premium  F      SI1     60.4    61    342  3.88  3.84  2.33  
## 7  0.31 Ideal    J      SI2     62.2    54    344  4.35  4.37  2.71  
## 8  0.2  Premium  E      SI2     60.2    62    345  3.79  3.75  2.27  
## 9  0.32 Premium  E      I1      60.9    58    345  4.38  4.42  2.68  
## 10 0.3  Ideal    I      SI2     62     54    348  4.31  4.34  2.68  
## # i 51,839 more rows
```

Filters with multiple conditions

```
diamonds %>%  
  filter(cut == "Very Good" | cut == "Fair")
```

```
## # A tibble: 13,692 x 10
```

```
##   carat cut      color clarity depth table price      x      y      z  
##   <dbl> <ord>    <ord> <ord>    <dbl> <dbl> <int> <dbl> <dbl> <dbl>  
## 1  0.24 Very Good J      VVS2    62.8    57   336   3.94   3.96   2.48  
## 2  0.24 Very Good I      VVS1    62.3    57   336   3.95   3.98   2.47  
## 3  0.26 Very Good H      SI1     61.9    55   337   4.07   4.11   2.53  
## 4  0.22 Fair      E      VS2     65.1    61   337   3.87   3.78   2.49  
## 5  0.23 Very Good H      VS1     59.4    61   338    4     4.05   2.39  
## 6  0.3   Very Good J      SI1     62.7    59   351   4.21   4.27   2.66  
## 7  0.23 Very Good E      VS2     63.8    55   352   3.85   3.92   2.48  
## 8  0.23 Very Good H      VS1     61     57   353   3.94   3.96   2.41  
## 9  0.31 Very Good J      SI1     59.4    62   353   4.39   4.43   2.62  
## 10 0.31 Very Good J      SI1     58.1    62   353   4.44   4.47   2.59  
## # i 13,682 more rows
```


Filter after select

This is an example of “a sequence of operations”.

```
diamonds %>%  
  select(price) %>%  
  filter(price > 335)
```

```
## # A tibble: 53,935 x 1  
##   price  
##   <int>  
## 1    336  
## 2    336  
## 3    337  
## 4    337  
## 5    338  
## 6    339  
## 7    340  
## 8    342  
## 9    344  
## 10   345  
## # i 53,925 more rows
```

Arrange

Use `arrange()` to order data.

```
diamonds %>%  
  arrange(price)
```

```
## # A tibble: 53,940 x 10  
##   carat cut      color clarity depth table price      x      y      z  
##   <dbl> <ord>    <ord> <ord>    <dbl> <dbl> <int> <dbl> <dbl> <dbl>  
## 1  0.23 Ideal    E     SI2     61.5   55   326   3.95   3.98   2.43  
## 2  0.21 Premium  E     SI1     59.8   61   326   3.89   3.84   2.31  
## 3  0.23 Good     E     VS1     56.9   65   327   4.05   4.07   2.31  
## 4  0.29 Premium  I     VS2     62.4   58   334   4.2    4.23   2.63  
## 5  0.31 Good     J     SI2     63.3   58   335   4.34   4.35   2.75  
## 6  0.24 Very Good J     VVS2    62.8   57   336   3.94   3.96   2.48  
## 7  0.24 Very Good I     VVS1    62.3   57   336   3.95   3.98   2.47  
## 8  0.26 Very Good H     SI1     61.9   55   337   4.07   4.11   2.53  
## 9  0.22 Fair     E     VS2     65.1   61   337   3.87   3.78   2.49  
## 10 0.23 Very Good H     VS1     59.4   61   338   4      4.05   2.39  
## # i 53,930 more rows
```

Arrange descending order

e.g. from the cheapest!

```
diamonds %>%  
  arrange(-price)
```

```
## # A tibble: 53,940 x 10  
##   carat cut      color clarity depth table price      x      y      z  
##   <dbl> <ord>    <ord> <ord>    <dbl> <dbl> <int> <dbl> <dbl> <dbl>  
## 1  2.29 Premium  I     VS2     60.8   60 18823  8.5   8.47  5.16  
## 2  2.00 Very Good G     SI1     63.5   56 18818  7.9   7.97  5.04  
## 3  1.51 Ideal    G     IF      61.7   55 18806  7.37  7.41  4.56  
## 4  2.07 Ideal    G     SI2     62.5   55 18804  8.2   8.13  5.11  
## 5  2.00 Very Good H     SI1     62.8   57 18803  7.95  8.00  5.01  
## 6  2.29 Premium  I     SI1     61.8   59 18797  8.52  8.45  5.24  
## 7  2.04 Premium  H     SI1     58.1   60 18795  8.37  8.28  4.84  
## 8  2.00 Premium  I     VS1     60.8   59 18795  8.13  8.02  4.91  
## 9  1.71 Premium  F     VS2     62.3   59 18791  7.57  7.53  4.70  
## 10 2.15 Ideal    G     SI2     62.6   54 18791  8.29  8.35  5.21  
## # i 53,930 more rows
```

Arrange by multiple conditions

```
diamonds %>%  
  arrange(price, cut)
```

```
## # A tibble: 53,940 x 10
```

##	carat	cut	color	clarity	depth	table	price	x	y	z
##	<dbl>	<ord>	<ord>	<ord>	<dbl>	<dbl>	<int>	<dbl>	<dbl>	<dbl>
## 1	0.21	Premium	E	SI1	59.8	61	326	3.89	3.84	2.31
## 2	0.23	Ideal	E	SI2	61.5	55	326	3.95	3.98	2.43
## 3	0.23	Good	E	VS1	56.9	65	327	4.05	4.07	2.31
## 4	0.29	Premium	I	VS2	62.4	58	334	4.2	4.23	2.63
## 5	0.31	Good	J	SI2	63.3	58	335	4.34	4.35	2.75
## 6	0.24	Very Good	J	VVS2	62.8	57	336	3.94	3.96	2.48
## 7	0.24	Very Good	I	VVS1	62.3	57	336	3.95	3.98	2.47
## 8	0.22	Fair	E	VS2	65.1	61	337	3.87	3.78	2.49
## 9	0.26	Very Good	H	SI1	61.9	55	337	4.07	4.11	2.53
## 10	0.23	Very Good	H	VS1	59.4	61	338	4	4.05	2.39

```
## # i 53,930 more rows
```

Filter, select, arrange

```
diamonds %>%  
  filter(table < 340) %>%  
  select(carat, cut, price) %>%  
  arrange(price, cut)
```

```
## # A tibble: 53,940 x 3  
##   carat cut      price  
##   <dbl> <ord>    <int>  
## 1  0.21 Premium     326  
## 2  0.23 Ideal      326  
## 3  0.23 Good       327  
## 4  0.29 Premium     334  
## 5  0.31 Good       335  
## 6  0.24 Very Good   336  
## 7  0.24 Very Good   336  
## 8  0.22 Fair       337  
## 9  0.26 Very Good   337  
## 10 0.23 Very Good   338  
## # i 53,930 more rows
```

Mutate

Create new variables using `mutate()`.

- Create a boolean variable, 0 = not affordable, 1 = affordable.

```
diamonds %>%  
  mutate(affordable = price < 400)
```

```
## # A tibble: 53,940 x 11  
##   carat cut      color clarity depth table price      x      y      z affordable  
##   <dbl> <ord>      <ord> <ord>    <dbl> <dbl> <int> <dbl> <dbl> <dbl> <lgl>  
## 1  0.23 Ideal      E      SI2     61.5    55   326   3.95   3.98   2.43 TRUE  
## 2  0.21 Premium    E      SI1     59.8    61   326   3.89   3.84   2.31 TRUE  
## 3  0.23 Good       E      VS1     56.9    65   327   4.05   4.07   2.31 TRUE  
## 4  0.29 Premium    I      VS2     62.4    58   334   4.2    4.23   2.63 TRUE  
## 5  0.31 Good       J      SI2     63.3    58   335   4.34   4.35   2.75 TRUE  
## 6  0.24 Very Good J      VVS2     62.8    57   336   3.94   3.96   2.48 TRUE  
## 7  0.24 Very Good I      VVS1     62.3    57   336   3.95   3.98   2.47 TRUE  
## 8  0.26 Very Good H      SI1     61.9    55   337   4.07   4.11   2.53 TRUE  
## 9  0.22 Fair       E      VS2     65.1    61   337   3.87   3.78   2.49 TRUE  
## 10 0.23 Very Good H      VS1     59.4    61   338   4      4.05   2.39 TRUE  
## # i 53,930 more rows
```

Mutate (cont'd)

- Create a variable containing string with `case_when()`:

```
diamonds %>%  
  mutate(affordable = case_when(price<400 ~ "affordable",  
                                TRUE ~ "not affordable"))
```

```
## # A tibble: 53,940 x 11  
##   carat cut      color clarity depth table price      x      y      z affordable  
##   <dbl> <ord>    <ord> <ord>    <dbl> <dbl> <int> <dbl> <dbl> <dbl> <chr>  
## 1  0.23 Ideal      E      SI2     61.5    55   326   3.95   3.98   2.43 affordable  
## 2  0.21 Premium   E      SI1     59.8    61   326   3.89   3.84   2.31 affordable  
## 3  0.23 Good      E      VS1     56.9    65   327   4.05   4.07   2.31 affordable  
## 4  0.29 Premium   I      VS2     62.4    58   334   4.2    4.23   2.63 affordable  
## 5  0.31 Good      J      SI2     63.3    58   335   4.34   4.35   2.75 affordable  
## 6  0.24 Very Good J      VVS2    62.8    57   336   3.94   3.96   2.48 affordable  
## 7  0.24 Very Good I      VVS1    62.3    57   336   3.95   3.98   2.47 affordable  
## 8  0.26 Very Good H      SI1     61.9    55   337   4.07   4.11   2.53 affordable  
## 9  0.22 Fair      E      VS2     65.1    61   337   3.87   3.78   2.49 affordable  
## 10 0.23 Very Good H      VS1     59.4    61   338   4      4.05   2.39 affordable  
## # i 53,930 more rows
```

Group by and Summarise

Use `group_by` and `summarise` to group variables:

```
diamonds %>%  
  group_by(cut) %>%  
  summarise(n = n())
```

```
## # A tibble: 5 x 2  
##   cut      n  
##   <ord>   <int>  
## 1 Fair    1610  
## 2 Good    4906  
## 3 Very Good 12082  
## 4 Premium 13791  
## 5 Ideal   21551
```


More examples

```
diamonds %>%  
  group_by(cut) %>%  
  summarise(n = n(), price_avg = mean(price))
```

```
## # A tibble: 5 x 3  
##   cut          n price_avg  
##   <ord>      <int>      <dbl>  
## 1 Fair        1610      4359.  
## 2 Good        4906      3929.  
## 3 Very Good  12082      3982.  
## 4 Premium    13791      4584.  
## 5 Ideal      21551      3458.
```

Proportions

```
diamonds %>%  
  group_by(cut) %>%  
  summarise(n = n(), price_avg = mean(price)) %>%  
  ungroup() %>%  
  mutate(prop = n/sum(n))
```

```
## # A tibble: 5 x 4  
##   cut          n price_avg  prop  
##   <ord>    <int>    <dbl> <dbl>  
## 1 Fair      1610     4359. 0.0298  
## 2 Good      4906     3929. 0.0910  
## 3 Very Good 12082     3982. 0.224  
## 4 Premium   13791     4584. 0.256  
## 5 Ideal     21551     3458. 0.400
```

With percentage

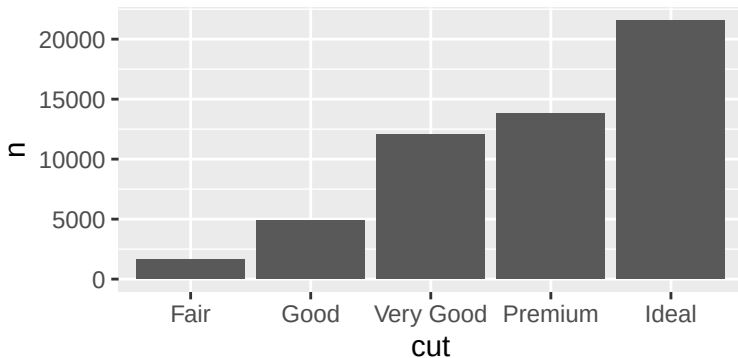
Use `scales::percent()` to add %.

```
diamonds %>%  
  group_by(cut) %>%  
  summarise(n = n(), price_avg = mean(price)) %>%  
  ungroup() %>%  
  mutate(prop = scales::percent(n/sum(n)))
```

```
## # A tibble: 5 x 4  
##   cut          n price_avg prop  
##   <ord>      <int>    <dbl> <chr>  
## 1 Fair       1610     4359. 3.0%  
## 2 Good       4906     3929. 9.1%  
## 3 Very Good 12082     3982. 22.4%  
## 4 Premium   13791     4584. 25.6%  
## 5 Ideal     21551     3458. 40.0%
```

Graphing after transformation

```
diamonds %>%  
  group_by(cut) %>%  
  summarise(n = n(), price_avg = mean(price)) %>%  
  ggplot() +  
  geom_bar(aes(x = cut, y = n), stat = "identity")
```



ggplot

Here we used functions from “ggplot2” package. Same pattern as “tidyverse”, but using “+” to connect.

How to write?

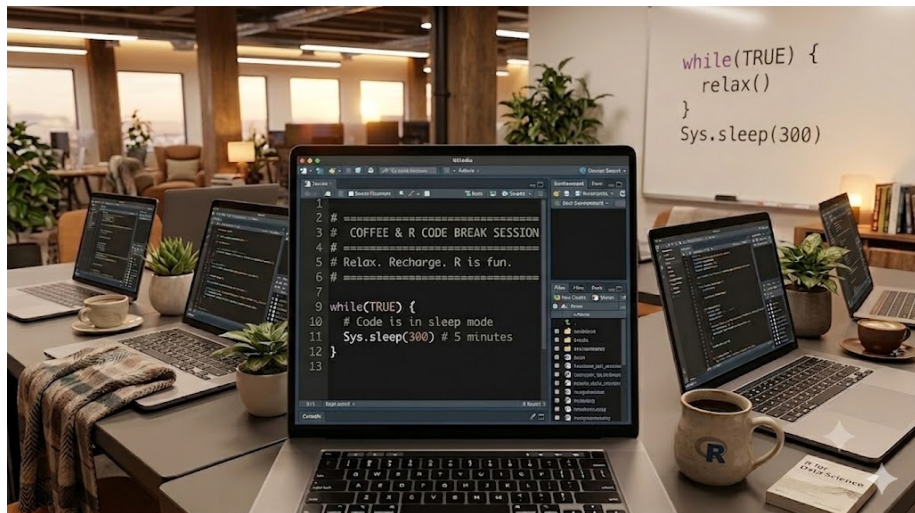
- Specify the data using `ggplot(data = diamonds)`
- Specify the x-/y-axis, `ggplot(data = diamonds, mapping = aes(x = cut))`
- Specify the types of plots with `geom`, e.g. `+ geom_bar()`

```
ggplot(data = diamonds) +  
  geom_bar(mapping = aes(x = cut))
```

More plots

- `geom_histogram()`, `geom_density()`, `geom_line()`, `geom_point()`
- `geom_facet()` generates subplots
- color package
 - RColorBrewer
 - ggsci
- formatting package
 - ggthemes

Break



Install and load package

From what we have learned so far, how to install package? What packages are we going to use?

Install and load package

From what we have learned so far, how to install package? What packages are we going to use?

```
install.packages("palmerpenguins")
```

```
library(tidyverse)  
library(palmerpenguins)
```

```
## Warning: package 'palmerpenguins' was built under R version
```

Skim the data

How many observations? How many variables? What type?

Skim the data

How many observations? How many variables? What type?

```
head(penguins)
```

```
## # A tibble: 6 x 8
##   species island    bill_length_mm bill_depth_mm flipper_le
##   <fct>    <fct>          <dbl>          <dbl>
## 1 Adelie  Torgersen         39.1           18.7
## 2 Adelie  Torgersen         39.5           17.4
## 3 Adelie  Torgersen         40.3           18
## 4 Adelie  Torgersen          NA           NA
## 5 Adelie  Torgersen         36.7           19.3
## 6 Adelie  Torgersen         39.3           20.6
## # i 2 more variables: sex <fct>, year <int>
```

Scatter plot

Consider a scatter plot of flipper length and body mass for `species = "Adelie"`

First let's prepare the data to plot (Hint: `filter`).

Scatter plot

Consider a scatter plot of flipper length and body mass for species = "Adelie"

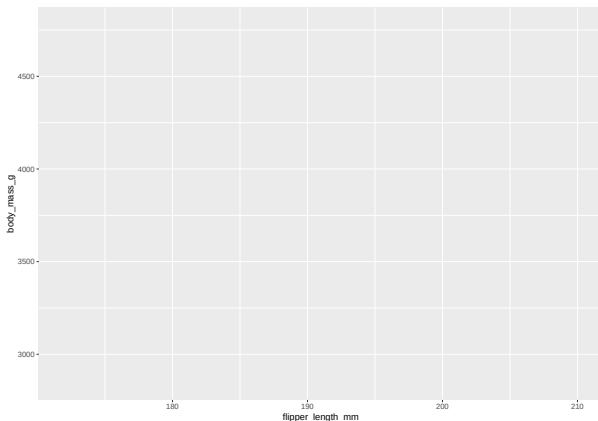
First let's prepare the data to plot (Hint: filter).

```
pdata <- penguins %>%  
  filter(species == "Adelie")
```

A blank canvas

`aes` stands for aesthetic and tells `ggplot` the main characteristics of your plot (x, y, and if the color or fill vary by group)

```
ggplot(data = pdata, aes(x = flipper_length_mm, y = body_mass_g))
```

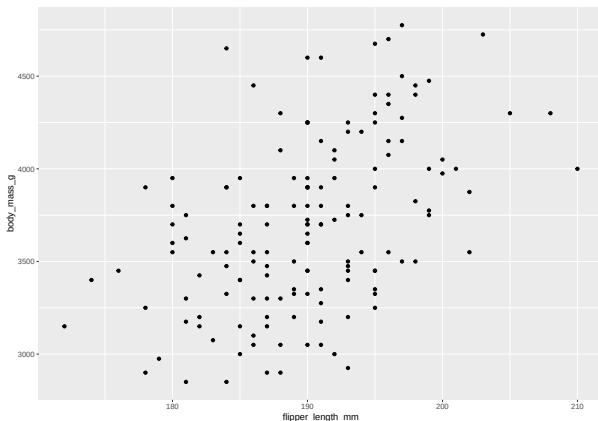


Add the points

Add layers with ggplot using the +

```
ggplot(data = pdata, aes(x = flipper_length_mm, y = body_mass_g)) +  
  geom_point()
```

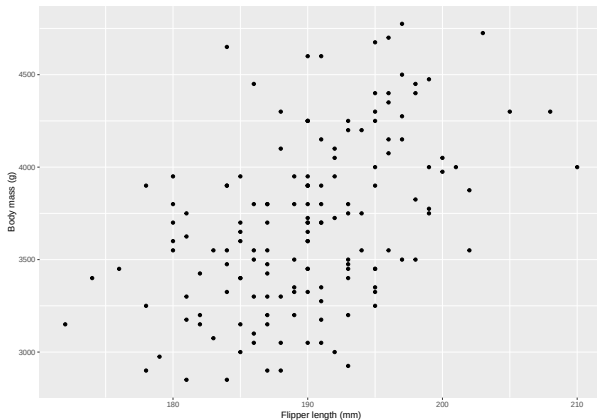
```
## Warning: Removed 1 row containing missing values or values outside the scale range  
## (`geom_point()`).
```



Tidy up labels

```
ggplot(data = pdata, aes(x = flipper_length_mm, y = body_mass_g)) +  
  geom_point() +  
  xlab("Flipper length (mm)") +  
  ylab("Body mass (g)")
```

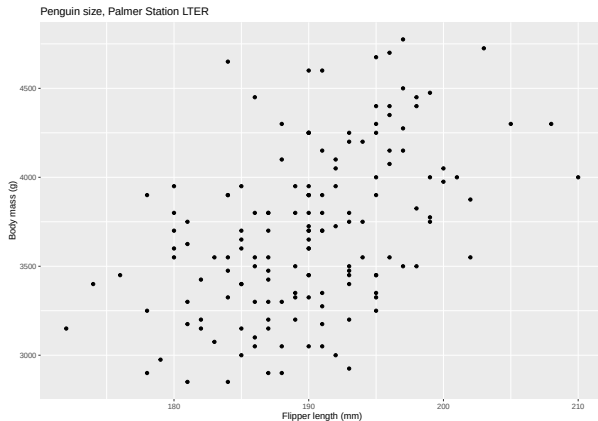
```
## Warning: Removed 1 row containing missing values or values outside the scale range  
## (`geom_point()`).
```



Add a title

```
ggplot(data = pdata, aes(x = flipper_length_mm, y = body_mass_g)) +  
  geom_point() +  
  xlab("Flipper length (mm)") +  
  ylab("Body mass (g)") +  
  labs(title = "Penguin size, Palmer Station LTER")
```

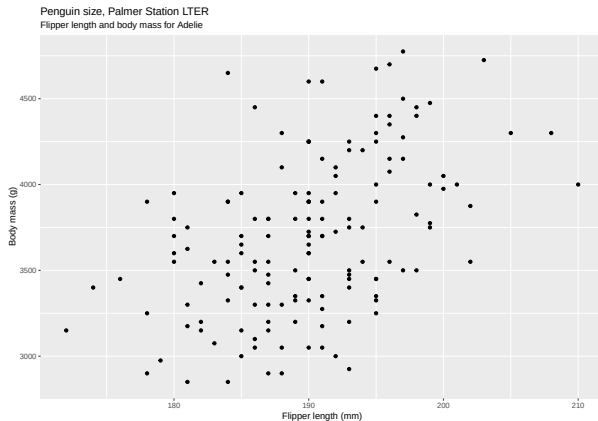
```
## Warning: Removed 1 row containing missing values or values outside the scale range  
## (`geom_point()`).
```



Subtitle

```
ggplot(data = pdata, aes(x = flipper_length_mm, y = body_mass_g)) +  
  geom_point() +  
  xlab("Flipper length (mm)") +  
  ylab("Body mass (g)") +  
  labs(title = "Penguin size, Palmer Station LTER",  
       subtitle = "Flipper length and body mass for Adelie")
```

```
## Warning: Removed 1 row containing missing values or values outside the scale range  
## (`geom_point()`).
```

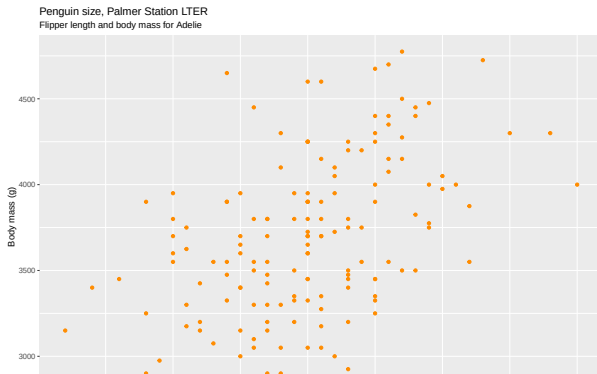


Change color of points

To see all colors, type `colors()`

```
ggplot(data = pdata, aes(x = flipper_length_mm, y = body_mass_g)) +  
  geom_point(color = "darkorange") +  
  xlab("Flipper length (mm)") +  
  ylab("Body mass (g)") +  
  labs(title = "Penguin size, Palmer Station LTER",  
        subtitle = "Flipper length and body mass for Adelie")
```

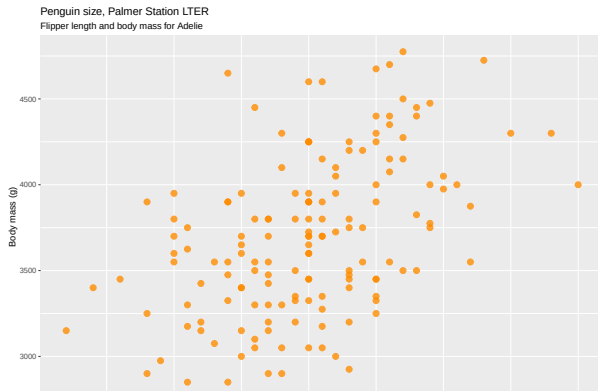
```
## Warning: Removed 1 row containing missing values or values outside the scale range  
## (`geom_point()`).
```



Size of points

```
ggplot(data = pdata, aes(x = flipper_length_mm, y = body_mass_g)) +  
  geom_point(color = "darkorange",  
            size = 3,  
            alpha = 0.8) +  
  xlab("Flipper length (mm)") +  
  ylab("Body mass (g)") +  
  labs(title = "Penguin size, Palmer Station LTER",  
       subtitle = "Flipper length and body mass for Adelie")
```

```
## Warning: Removed 1 row containing missing values or values outside the scale range  
## (`geom_point()`).
```

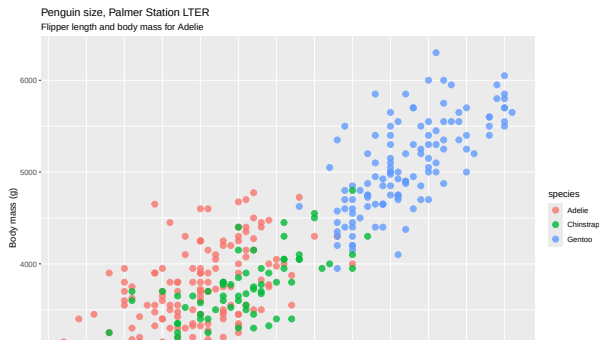


Coloring by group

Now use the whole data penguins. Specifying the color in `aes()` because it **depends on the data**.

```
ggplot(data = penguins, aes(x = flipper_length_mm, y = body_mass_g, color = species)) +  
  geom_point(size = 3, alpha = 0.8) +  
  xlab("Flipper length (mm)") +  
  ylab("Body mass (g)") +  
  labs(title = "Penguin size, Palmer Station LTER",  
        subtitle = "Flipper length and body mass for Adelie")
```

```
## Warning: Removed 2 rows containing missing values or values outside the scale range  
## (`geom_point()`).
```

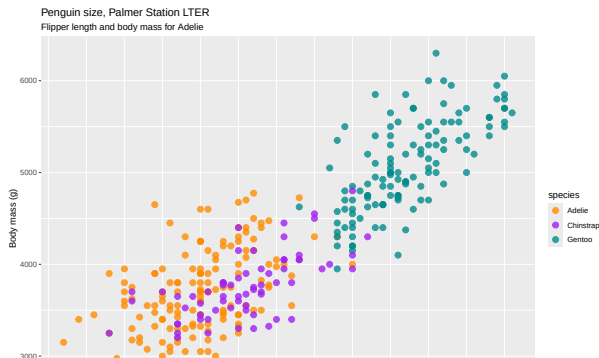


Change the color manually by group

Specifying the color in `aes()` because it **depends on the data**.

```
ggplot(data = penguins, aes(x = flipper_length_mm, y = body_mass_g, color = species)) +  
  geom_point(size = 3, alpha = 0.8) +  
  scale_color_manual(values = c("darkorange", "purple", "cyan4")) +  
  xlab("Flipper length (mm)") +  
  ylab("Body mass (g)") +  
  labs(title = "Penguin size, Palmer Station LTER",  
       subtitle = "Flipper length and body mass for Adelie")
```

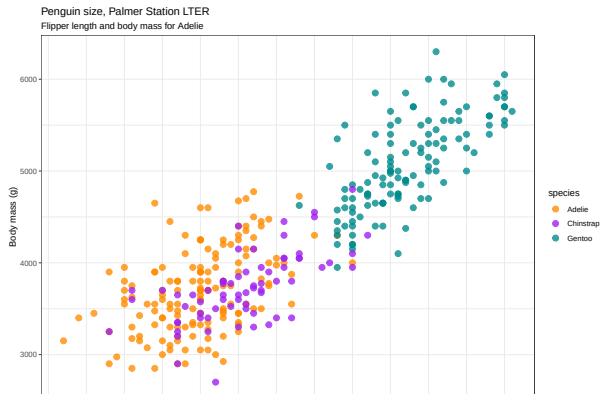
```
## Warning: Removed 2 rows containing missing values or values outside the scale range  
## (`geom_point()`).
```



Change the theme

```
ggplot(data = penguins, aes(x = flipper_length_mm, y = body_mass_g, color = species)) +  
  geom_point(size = 3, alpha = 0.8) +  
  scale_color_manual(values = c("darkorange", "purple", "cyan4")) +  
  xlab("Flipper length (mm)") +  
  ylab("Body mass (g)") +  
  labs(title = "Penguin size, Palmer Station LTER",  
       subtitle = "Flipper length and body mass for Adelie") +  
  theme_bw()
```

```
## Warning: Removed 2 rows containing missing values or values outside the scale range  
## (`geom_point()`).
```

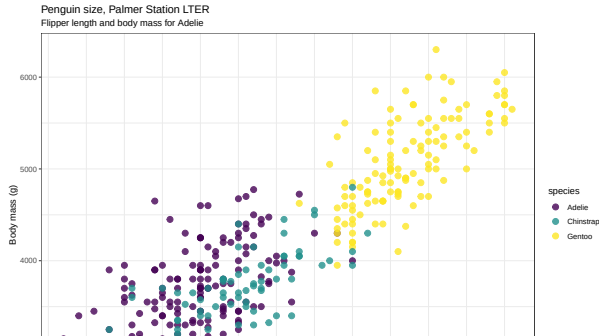


Change the color scheme

Optional: try ggsci functions.

```
ggplot(data = penguins, aes(x = flipper_length_mm, y = body_mass_g, color = species)) +  
  geom_point(size = 3, alpha = 0.8) +  
  xlab("Flipper length (mm)") +  
  ylab("Body mass (g)") +  
  labs(title = "Penguin size, Palmer Station LTER",  
       subtitle = "Flipper length and body mass for Adelie") +  
  theme_bw() +  
  scale_color_viridis_d()
```

```
## Warning: Removed 2 rows containing missing values or values outside the scale range  
## (`geom_point()`).
```



Other plot types?

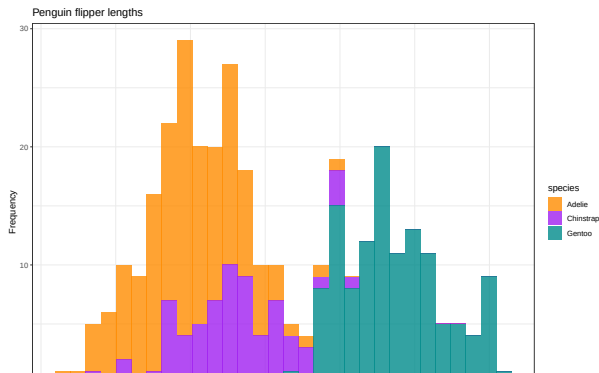
Other types are also available, e.g. histograms, bar charts, box plots, line graphs and scatter plots.

Histograms

```
ggplot(data = penguins, aes(x = flipper_length_mm, fill = species)) +  
  geom_histogram(alpha = 0.8) +  
  scale_fill_manual(values = c("darkorange", "purple", "cyan4")) +  
  xlab("Flipper length (mm)") +  
  ylab("Frequency") +  
  labs(title = "Penguin flipper lengths") +  
  theme_bw()
```

```
## `stat_bin()` using `bins = 30`. Pick better value with `binwidth`.
```

```
## Warning: Removed 2 rows containing non-finite outside the scale range  
## (`stat_bin()`).
```

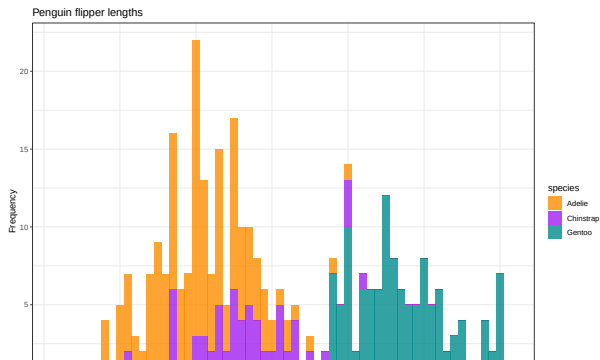


Histogram

Try `binwidth =` and `bins ==`

```
ggplot(data = penguins, aes(x = flipper_length_mm, fill = species)) +  
  geom_histogram(binwidth = 1, alpha = 0.8) +  
  scale_fill_manual(values = c("darkorange", "purple", "cyan4")) +  
  xlab("Flipper length (mm)") +  
  ylab("Frequency") +  
  labs(title = "Penguin flipper lengths") +  
  theme_bw()
```

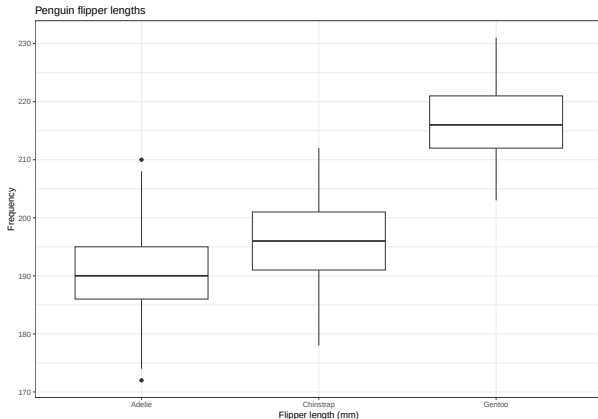
```
## Warning: Removed 2 rows containing non-finite outside the scale range  
## (`stat_bin()`).
```



Boxplots

```
ggplot(data = penguins, aes(x = species, y = flipper_length_mm)) +  
  geom_boxplot(show.legend = FALSE) +  
  xlab("Flipper length (mm)") +  
  ylab("Frequency") +  
  labs(title = "Penguin flipper lengths") +  
  theme_bw()
```

```
## Warning: Removed 2 rows containing non-finite outside the scale range  
## (`stat_boxplot()`).
```



Barcharts

First, try to summarize the penguin data by species and returns the proportion of each penguin types.

Barcharts

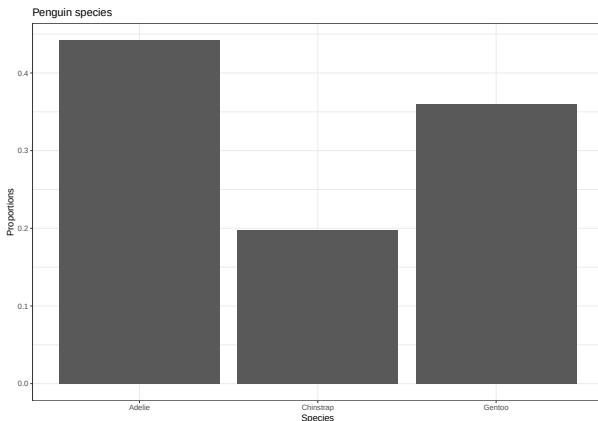
First, try to summarize the penguin data by species and returns the proportion of each penguin types.

```
gplot <- penguins %>%  
  group_by(species) %>%  
  tally() %>%  
  mutate(prop = n / sum(n))  
gplot
```

```
## # A tibble: 3 x 3  
##   species      n prop  
##   <fct>    <int> <dbl>  
## 1 Adelie    152 0.442  
## 2 Chinstrap  68 0.198  
## 3 Gentoo   124 0.360
```

Barcharts

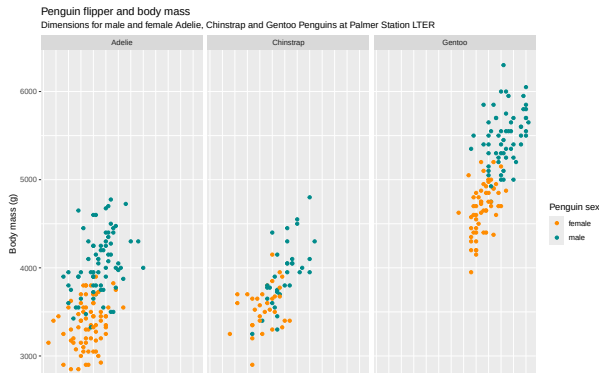
```
ggplot(data = gplot, aes(x = species, y = prop)) +  
  geom_bar(stat = "identity") +  
  xlab("Species") +  
  ylab("Proportions") +  
  labs(title = "Penguin species") +  
  theme_bw()
```



Faceting

```
ggplot(penguins, aes(x = flipper_length_mm, y = body_mass_g)) +  
  geom_point(aes(color = sex)) +  
  scale_color_manual(values = c("darkorange", "cyan4"), na.translate = FALSE) +  
  labs(title = "Penguin flipper and body mass",  
       subtitle = "Dimensions for male and female Adelie, Chinstrap and Gentoo Penguins at Palmer Station LTER",  
       x = "Flipper length (mm)",  
       y = "Body mass (g)",  
       color = "Penguin sex") +  
  facet_wrap(~species)
```

```
## Warning: Removed 11 rows containing missing values or values outside the scale range  
## (`geom_point()`).
```



Git + Github

What is Git?

- A control system to manage projects
- Good for tracking history

Git + Github

What is Git?

- A control system to manage projects
- Good for tracking history

What about Github?

- Cloud-based service for managing Git repositories
- Useful for teamwork
- Just like “Dropbox”

Why Git + Github?

- You can undo anything
- You won't need to keep undo-ing things (merge/load difference)
- You can identify exactly when and where changes were made
- Teamwork

Some github terminology

- User: A Github account for you (e.g., jules32).
- Organization: The Github account for one or more user (e.g., datacarpentry).
- Repository: A folder within the organization that includes files dedicated to a project.
- Local Github: Copies of Github files located your computer.
- Remote Github: Github files located on the <https://github.com> website.

Basic Git commands and workflow

When you are working on a your local machine you typically get started by:

- `git clone`: Cloning a remote repository to work on locally. This is a way to work with an ongoing project or edit someone else's project that is available remotely (aka on GitHub).

From there, the typical workflow involves:

- `git add`: Adding files to your repo
- `git commit`: Committing changes you have made
- `git push`: Pushing changes to a remote repository (aka GitHub)

For a collaborative project, or work between desktop and personal laptop, you would use the following first before `git add`

- `git pull`: Pulling changes from a remote repository

Illustration diagram

REMOTE (aka Github website)



Clone (i.e., copy)
repository to your
computer (a one
time event)

**Pull remote
changes**

**Push local
changes**

Almost all work
is done 'locally'



LOCAL

Let's Git!

Download Github and set up your github profile.

- Github Desktop, a GUI for using Github [\[link\]](#)
- (Optional) Learn how to use command line for Github management
 - Command line tutorial [\[link\]](#)

Exercises

Available on course website.